

Influence of fire and harvest severity on understory plant communities

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ABSTRACT

In the managed portion of the boreal forest of eastern North America, logging has replaced fire as the most important disturbance agent. There, a large proportion of timber is harvested in forests susceptible of accumulating a thick *Sphagnum* layer that decreases forest productivity, a process called paludification. In such a context, understanding how disturbance type and severity of soil disturbances may affect post-disturbance microhabitat characteristics and understory community composition is critical for forest management. Different management techniques have been used such as careful logging and clearcutting, as well as winter and summer harvests, with various impacts on soils and forest regeneration. In the current study, we used 55 study sites representing a gradient of soil disturbance severity by harvesting (winter and summer) and fire (low and high severity) to compare their impacts on understory plant communities in the Clay Belt area of eastern Canada. At each site, understory community composition (vascular plants, bryophytes, and lichens) was assessed. We found a strong response of communities to overall severity as represented by disturbance type (careful logging, clearcutting, and fire), but little impact of fine scale disturbance severity (winter vs. summer, low vs. high severity disturbance) within each type of disturbance. Differences in community composition were reflected in the abundance of the various plant functional types, with invaders being more common in harvested sites, endurers being common in all disturbances except high severity fires, and avoiders being more common in older sites. Understory communities in harvested sites (< 40 years old) were similar to communities typical of old sites originating from natural wildfire disturbances (75–100 years old low severity fires or 200 years old high severity fires) in terms of composition, but also *Sphagnum* spp. abundance. In order to maintain long-term forest productivity and manage forests in ways that more closely reproduce post-fire conditions, logging operations should aim at increasing soil disturbances, for example by using prescribed burns, in the Clay Belt area of eastern North America.

1. Introduction

Stand-replacing wildfires are the primary disturbances driving natural dynamics in boreal forested ecosystems (Bergeron et al., 2001; Johnstone et al., 2010). Fire characteristics and postfire succession strongly influence stand composition and structure, as well as understory community composition (Johnstone and Chapin, 2006; Lecomte et al., 2006a; Hart and Chen, 2008). However, in much of the boreal forest of North America, logging and forest management have replaced fire as the most important disturbance agent (Burton et al., 2010), sometimes affecting larger areas than natural fire disturbances (Bergeron et al., 2006; Drapeau et al., 2009). With growing concerns regarding a sustainable use of forest resources and the maintenance of ecosystem functions (CCFM, 1995), various forest management techniques have been applied, some of which aim at protecting the soil and tree regeneration or, alternatively, at more closely

imitating fire by increasing soil disturbances (OMNR, 1997; Gauthier et al., 2009). Post-disturbance microhabitat characteristics and temporal changes in microhabitat availability are expected to differ between fire and clear-cut logging and according to the severity of soil disturbances that expose mineral soil as both disturbance types completely open the canopy (Nguyen-Xuan et al., 2000; Harper et al., 2004). Mineral soil is a high-quality seedbed, and for trees and other plants, its availability following high severity fires that consume much of the soil organic layer (SOL), has been linked to both different understory community compositions but also to increased stand productivity when compared to low severity fires (Johnstone and Chapin, 2006; Greene et al., 2007; Simard et al., 2007). Despite the importance of understory communities on ecosystem processes in boreal forests (Fenton and Bergeron, 2006), it is still unclear how fire and logging of various severities affect community composition of vascular plants, bryophytes, and lichen communities.

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Typically, studies comparing the effects of wildfires and logging on understory plant communities do not take disturbance severity into account, tend to compare wildfire with one or two logging treatments to identify which treatments most resemble wildfire (e.g., Nguyen-Xuan et al., 2000; Hart and Chen, 2008), and often focus on one taxonomic group, such as bryophytes (Paquette et al., 2016), or lichens (Lafleur et al., 2016b). Stand-replacing fires open up the canopy, kill the understory vegetation, burn the forest floor and downed wood debris, remove or reduce the depth of the soil organic layer (SOL), and may expose the mineral soil over a relatively large proportion of the burned area (Viereck, 1983; Miyanishi and Johnson, 2002). Early postfire understory communities are comprised of broad spectrum or shade intolerant species that resprout from surviving underground parts and species that have efficient long-distance dispersal strategies (Dyrness, 1973). Conversely, mechanical disturbances associated with logging do not completely eliminate understory vegetation, remove forest floor and downed woody debris, or expose mineral soil over large areas (Nguyen-Xuan et al., 2000; Lafleur et al., 2018). Post-logging understory communities are more similar to pre-logging communities, with the exception of late successional shade intolerant species that do not survive stand opening induced by logging (De Grandpré et al., 2003; Faivre et al., 2016). With time, initial differences in understory composition between logged and burned sites tend to dissipate (Reich et al., 2001; Rees and Juday, 2002; Rydgren et al., 2004). This convergence likely indicates that boreal understory communities are dominated by resilient species that possess traits and reproductive strategies than enable them to survive through or recolonize following disturbance (Rowe, 1983a,b; Reich et al., 2012; Hollingsworth et al., 2013).

Two main harvesting methods are typically used in the boreal forest, i.e. clear-cut (CC) harvesting that has been the main method used until the 1990s and careful logging with protection of regeneration and soils (CL) that is the currently recommended method in eastern Canada (OMNR, 1997). CC leads to higher soil disturbance, as the machinery mixes the organic and mineral layers of the soil, which increases decomposition rates, nutrient availability, and pH, along with removing much of the understory vegetation (Marshall, 2000; Lafleur et al., 2010, 2018). In contrast, CL has a limited impact on soils and understories, since machinery disturbance is limited to specific trails and soils, understory plants, and trees < 9 cm diameter at breast height are left untouched in-between trails (Lafleur et al., 2010; Faivre et al., 2016). In addition, summer harvests lead to higher soil disturbances than winter harvests, when the frozen ground and snow cover protect the soil and vegetation (Lafleur et al., 2010). Indeed, recent studies comparing fire and logging treatments over 40 years showed that disturbance severity explained stand productivity: CC, a high-severity disturbance, was able to produce stands as productive as high-severity fires, while stand productivity was lower following CL and low-severity fires (Simard et al., 2007; Lafleur et al., 2010, 2018). Given the various impacts of fire and logging disturbances on soils, the recovery of understory plant communities may depend on the different life-history strategies of boreal plants, as well as historical successional patterns (Lafleur et al., 2010, 2016a, 2018).

Within the boreal forest, differences in length of the fire cycle lead to differential successional patterns and relative importance of environmental conditions. In the southern portion of the boreal forest, long fire cycles allow a directional succession from deciduous-dominated to coniferous-dominated forests and variations in light availability through time since disturbance is a major factor controlling understory community composition (Bergeron and Dubuc, 1989; De Grandpré et al., 1993). After canopy closure, many shade-intolerant pioneer species disappear, and are replaced by late-succession shade tolerant species (De Grandpré et al., 2003). In contrary, shorter fire cycles typical of coniferous boreal forests dominated by black spruce (*Picea mariana* (Mill.) B.S.P.) lead to structural succession with little change in canopy species composition (Harper et al., 2005). There, light availability tends to remain relatively high throughout succession and

species found there can usually withstand a wide range of environmental conditions along with recolonizing rapidly following disturbances from underground rhizomes (Schimmel and Granström, 1996; Harper et al., 2005; Fenton and Bergeron, 2013). In these stands, however, bryophytes do go through succession, with feather mosses being gradually replaced by *Sphagnum* along with a decrease in water table depth (Fenton et al., 2005; Fenton and Bergeron, 2006). In the Clay Belt area of eastern Canada, a large proportion of commercial timber is harvested in black spruce (*Picea mariana*) stands liable to paludification (Lafleur et al., 2018), i.e., the accumulation of a thick *Sphagnum* layer that decreases forest productivity in the long-term given a combination of cold climate and poorly drained soils (Fenton et al., 2005; Lecomte et al., 2005; Simard et al., 2007). In such a context, repercussions of soil disturbance severity on post-disturbance understory community regeneration could be particularly critical and outweigh changes in light availability.

Rowe (1983a,b) identified five reproductive strategies in plants, which illustrate the various adaptations of boreal plants to maintain their presence in the landscape despite the recurrence of disturbances. Among these strategies, three are propagule-based: *invaders* are typically short-lived shade-intolerant pioneer species with easily dispersed propagules (e.g., *Chamerion angustifolium* (L.) Scop.), *evaders* are pioneer species that rely on seed banks in the soil or in the canopy (e.g., *Viburnum cassinoides* (L.) Torr. & A. Gray), whereas *avoiders* are species that reproduce through seed dispersal but typically only establish in late successional forests (e.g., *Coptis groenlandica* (O.F. Müll.) Fernald). The other strategies are based on vegetative reproduction: *endurers* can resprout from underground organs that survive the fire (e.g. *Vaccinium angustifolium* Ait. or *Rhododendron groenlandicum* (Oeder) Kron & Judd) and *resisters* are species whose aboveground parts survive low-severity fires (e.g. *Pinus banksiana* Lamb. or *Eriophorum vaginatum* L.). While it is recognized that in fire-dominated boreal landscapes the abundance of each of these functional groups is determined by the fire frequency (i.e., invaders are more abundant where fires are frequent, whereas avoiders dominate where fire are relatively rare) (De Grandpré et al., 2003), the effect of different logging methods and seasons on these groups is less well understood.

In the current study, we used sites representing a gradient of soil disturbance severity by harvesting and fire to compare their impacts on understory plant communities (vascular plants, bryophytes, and lichens) in the Clay Belt area of eastern Canada. The general objective was to understand how forest harvest affects dynamics of the understory plant communities in black spruce forests with a potential for paludification. Specifically, we addressed the following questions: (1) What are the differences in vascular and non-vascular understory plant community composition among forest harvest types (CC vs. CL) and season (summer vs. winter)? (2) How do plant functional types respond to those same harvest types? (3) Do high and low severity harvest methods result in post-disturbance understory plant communities that are similar to high and low severity fires? We posit the following hypotheses. For both fire and harvest, increasing soil disturbance severity should result in increased cover of invader species, and a decrease in avoider species, especially in the years following disturbance. Likewise, endurers should be more abundant where soil disturbances are minimal. For both fire and harvest, the differences in understory plant community become less pronounced over time (i.e. time since disturbance). Comparing harvest and fire, higher severity soil disturbance harvest methods (i.e. summer CC) should result in patterns of community change that are more similar to high severity fire than CL methods.

2. Methods

2.1. Study area

Located in the Canadian Boreal Shield Ecozone, the Clay Belt of northeastern Ontario and northwestern Québec (Fig. 1) is a large

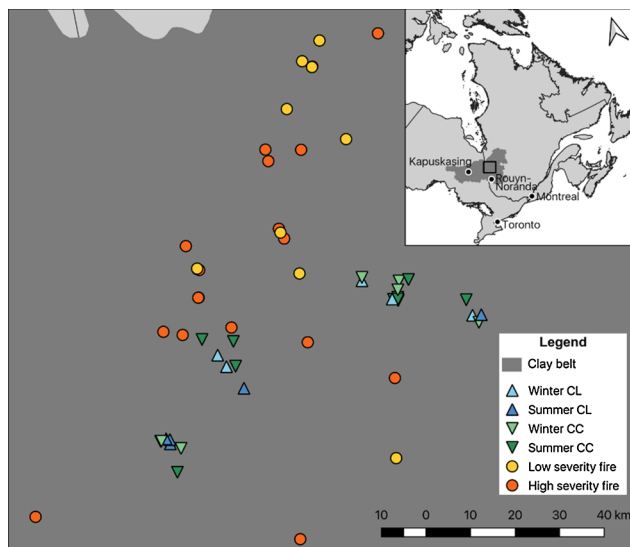


Fig. 1. Location of the study sites in the Clay Belt of eastern Canada. Site symbology represents disturbance types: careful logging (CL) as blue triangles, clear-cuts as green upside-down triangles, and fires as orange circles. Color shade indicates disturbance severity: lighter colors indicate lower severity fires and winter harvests and darker colors indicate high severity fires and summer harvests. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

physiographic region (125,000 km²) characterized by clay deposits (Vincent and Hardy, 1977). While the southern part of the study area is covered by a thick (> 10 m) layer of glaciolacustrine clay and silt deposited by the glacial Lake Ojibway, the northern part is covered by the Cochrane till, a compact till made up of a mixture of clay and gravel, created by a southward ice flow approximately 8000 years BP (Veillette, 1994). Soils of the study area are mostly classified as Gleysols and Luvisols (Soil Classification Working Group, 1998). Organic deposits (i.e. a surficial deposit consisting of a SOL > 30 cm thick) are found in many locations in both the southern and northern parts of the study area. According to the local weather station (Joutel [49°28'N; 78°18'W], Québec) from 1981 to 2010, the average annual temperature was 0.0 °C, the average annual precipitation was 909 mm (with 30% falling during the growing season), and the average number of growing degree-days (> 5 °C) was 1240 (Environment Canada, 2016). The frost-free season lasts about 100 days, with frost occurring occasionally during the growing season.

The study area is dominated by black spruce-feathermoss forests that vary in density and height. Occasional stands of jack pine (*P. banksiana*) and aspen (*Populus tremuloides* Michx.) occur and are often mixed with black spruce (Gauthier et al., 2000; Taylor, 2000; Harper et al., 2003). Larch (*Larix laricina* [Du Roi] K. Koch), balsam fir (*Abies balsamea* [L.] Mill.), and paper birch (*Betula papyrifera* Marsh.) are minor components of the forest matrix. Ericaceous shrubs such as *R. groenlandicum*, *Kalmia angustifolia* L., and blueberry (*Vaccinium myrtilloides* Michx. and *V. angustifolium*) dominate the understory, while the forest floor is dominated by *Sphagnum* spp. (e.g., *Sphagnum recurvum* P. Beauv. sensu lato, *Sphagnum capillifolium* [Ehrh.] Hedw., *Sphagnum fuscum* [Schimp.] Klinggr., *Sphagnum girgensohnii* Russ., and *Sphagnum magellanicum* Brid.) and feathermosses (e.g., *Pleurozium schreberi* [Brid.] Mitt., *Hylocomium splendens* [Hedw.] Schimp., and *Ptilium crista-castrensis* [Hewd.] De Not.). Paludification is a common phenomenon in the study area (Taylor et al., 1987), with *Sphagnum* spp. invading the forest floor after about 100–150 years following fire (Fenton et al., 2005).

2.2. Site selection

This study took advantage of sites that had been harvested under different forest management regimes (Lafleur et al., 2010) and sites that

experienced low (residual organic matter [ROM] > 5 cm) or high (ROM < 5 cm) severity fires (Lecomte et al., 2005, 2006a) to examine the long-term effects of different silvicultural practices and wildfires on understory community composition. Forest maps produced by Tembec, the major forest company in the region, were used to identify 27 paludified stands, which are stands with open multi-layer canopy with soil organic layer > 25 cm thick, that had been harvested by CL in winter (5 replicates; time since disturbance [TSD] = 11–18 years) or in summer (5 replicates; TSD = 16–18 years), and stands that had been CC in winter (8 replicates; TSD = 15–24 years) or in summer (9 replicates; TSD = 16–32 years) (Appendix A). Burned sites were selected by overlaying fire reconstruction maps (Bergeron et al., 2004) onto forest inventories produced by the Ministère des Ressources Naturelles et de la Faune (MRNF, 2009). A date since last fire was assigned and validated using the oldest age determined for three dominant trees per site (Lecomte et al., 2005, 2006a). The fire reconstruction maps were developed by dating fire scars and post-fire tree cohorts, and by using archives and aerial photographs for recent fires (see Bergeron et al., 2004 for further details). For both fire and harvest, only sites with a slight slope (< 4%) and a clay mineral soil were selected. Using these methods, we identified low (10 replicates; TSD = 38–169 years) and high (18 replicates; TSD = 52–353 years) severity fire sites (Appendix A; Fig. 1). In the study area, post-disturbance succession is dominated by black spruce and there is little change in canopy species composition (De Grandpré et al., 2003; Harper et al., 2005), which reduces the impact of having different age ranges for fire and harvest sites.

2.3. Data collection

At each site, the effects of harvest and fire on understory vegetation (i.e., vascular and non-vascular species) were assessed using slightly different methodologies for each disturbance type. Surveys were conducted during the summers of 2006, 2007, and 2008 for the harvest sites (Lafleur et al., 2010), and of 2000, 2001, and 2002 for the fire sites (Lecomte et al., 2005, 2006a) (Appendix A). In harvested sites, three 400 m² circular plots were placed 50 m apart, in which we counted the total number of black spruce stems as well as their height (Lafleur et al., 2010). Depth of the fibric, mesic, and humic organic layers (cm) were measured below three randomly selected trees within each plot, and samples of the organic layer were collected at a depth of 10–20 cm. Using these soil samples, we determined the chemical properties of soils at each harvested site: pH, carbon [C] to nitrogen [N] ratio, phosphorus (P) and base cations concentration. We also determined cation exchange capacity (CEC; defined as the sum of exchangeable cations, i.e. Ca, Mg, Na, K, Mn, Fe and Al), and base saturation (the sum of Ca, Mg, K, Na divided by CEC) (see Lafleur et al., 2010 for details on soil analyses). Within each fire site, one representative 10 × 10 m² plot was placed (Lecomte et al., 2005, 2006a). Within this plot, density, height, and composition of the post-fire canopy were measured. Depth of the ROM and total organic layer depth were measured on two to four organic layer monoliths dug down to the mineral soil (see Lecomte et al., 2005, 2006a for further details). In harvest sites, understory vegetation was visually surveyed in three 16 m² plots, while it was surveyed in ten 1 m² square plots for fire sites (Lecomte et al., 2005, 2006a). In both site types, all vascular plants, bryophytes, and lichens were identified at the species level whenever possible (including *Sphagnum* for harvest data, but not for fire data) or at the genus level (Appendix B). Percent ground cover of each taxa was recorded; taxa were evaluated individually so that total cover can exceed 100%. Each vascular and non-vascular taxa was also classified in one of five plant response strategies to disturbance, i.e. avoider, endurer, invader, evaders, and resisters (Appendix B; Flinn and Wein, 1977; Rowe, 1983a,b; Viereck, 1983; Taylor et al., 1987; Brumelis and Carleton, 1989; Wang and Kembell, 2005). All data were averaged per site prior to analysis.

2.4. Data analyses

2.4.1. Response of vegetation to harvest types and seasons

We conducted PerMANOVAs (multivariate analyses of variance [ANOVA] by permutations) with 5000 permutations to assess the difference according to the interaction among harvest type and season, along with time since disturbance (Anderson and Legendre, 1999). These analyses were conducted using the *adonis* function in the package *vegan* (Oksanen et al., 2017) in R (R Development Core Team, 2016) with vegetation communities converted into a Bray-Curtis distance matrix, as it is an appropriate distance measure for community data (Anderson and Legendre, 1999). Significant differences were followed by a pairwise Tukey HSD posthoc test (Martinez Arbizu, 2017).

These results were supported by a visual analysis of taxa composition using a nonmetric multidimensional scaling (NMDS) ordination conducted on taxa data from the 27 sites using the Bray-Curtis distance, 20 independent runs, and 200 iterations per run (McCune et al., 2002). We compared the cover of each of the three main functional types found in our species data (avoider, invader, and endurer) according to harvest type (CL vs. CC), season (winter vs. summer), their interaction, and time since disturbance (years) using linear models with a type III sum of squares. Non-significant interactions were removed from the final models (Engqvist, 2005). No statistical analyses were run on the evader and resister functional types, since they had a low occurrence and abundance at our sites, as well as very skewed data distribution. We then conducted individual linear models for all taxa that were present in at least 25% of the sites. This threshold was selected to avoid rare species with very skewed data distributions. Taxa cover was square root transformed prior to analysis to comply with assumptions of normality and homoscedasticity.

2.4.2. Similarity between harvest and fire post-disturbance communities

In order to assess if high and low severity harvest methods result in post-disturbance understory plant communities that are similar to high and low severity fires, we first merged the data from the harvest and fire datasets. Taxonomy resolution was adjusted to the lowest resolution to avoid issues associated with species detection. We conducted perMANOVAs on the community level data with disturbance types converted into an ordered factor (1–6) according to their severity or degree of soil disturbance: winter CL, summer CL, winter CC, summer CC, low severity fire and high severity fire. Time since disturbance was also included as a factor and illustrated the results using NMDS ordinations using the same methodology as for the response of vegetation to harvest type and season. Indicator taxa were identified for each harvest type and season combination using the *indval* function (labdsv; Dufrêne and Legendre, 1997; Roberts, 2007), based only species' specificity (abundance) and fidelity (presence) to a group (Legendre and Legendre, 2012). Cover of the three main functional types (avoider, invader, and endurer) were compared among disturbances severities and time since disturbance using linear models with a type III sum of squares. Given the much larger span of time since disturbance with the two datasets merged compared to the harvest data only and the somewhat older age of the fire sites compared to the harvest sites, we tested for the interaction between time since disturbance and disturbance severity. Time since disturbance was log transformed given the large number of younger sites and low number of older sites, and invader cover was square root transformed to comply with linear model assumptions.

2.4.3. Potential for paludification according to soil disturbance severity

We expect that higher severity harvest disturbances (CC) will have recovery patterns more similar to high severity fires and lower severity harvest disturbances (CL) to be more similar to low severity fires. We used the cover of *Sphagnum* as a proxy for paludification potential, since this genus plays a key role in this process. Differences in *Sphagnum* cover according to time since disturbance and fire severity, as well as differences among harvest types and seasons were assessed using linear models. Ericaceous shrubs are known to facilitate the vertical growth of

Sphagnum at the detriment of other ground covering bryophytes and lichens (Malmer et al., 2003; Fenton and Bergeron, 2006). A Pearson correlation was used to compare ericaceous shrub and *Sphagnum* spp. cover across all disturbance types.

3. Results

3.1. Functional types and growth form in plant communities

All the species, genera and families identified in the harvest and fire datasets were attributed to one of the five functional types (Appendix B). Thirty-four taxa were classified as avoiders: 19 bryophytes, 4 lichens and 11 vascular plants, mostly forbs and dwarf shrubs. Endurers encompassed 24 taxa, all of which were vascular plants, mostly shrubs and forbs. We classified 17 taxa as invaders: 5 bryophytes, 1 lichen and 11 vascular plants (forbs, graminoids and shrubs). Evaders comprised 7 taxa of vascular plants. Lastly, only one resister was identified: *Eriophorum* spp.

3.2. Community responses to harvest operations

We found a significant interaction between harvest type and season on the understory community composition of vascular plants ($p = 0.0326$; Table 1a), bryophytes and lichens ($p = 0.0296$; Table 1b) and the total vegetation ($p = 0.0258$; Appendix C). Time since disturbance had a negligible impact on vegetation community composition of the three subsets of data ($p > 0.05$; Table 1; Appendix C). Following pairwise posthoc tests ($p < 0.05$), we found that the only differences among treatments were between summer CL and summer CC, with the two winter disturbances showing intermediate community compositions (Fig. 2a and d). Vascular plants from the avoider and the endurer functional types were more associated with CL harvest, while CC harvests were more associated with plants from the invader functional type, particularly when harvest took place in the winter (Fig. 2b). Most bryophytes and lichens were from the avoider group, and no clear trend in the distribution of functional types was detected according to harvest type or season (Fig. 2c and d).

Avoider cover was higher in CL than in CC sites ($p = 0.0213$, Table 2a; Fig. 3a). There was a non-significant trend (interaction $p = 0.0958$) for sites with intermediate disturbance severity to have a slightly higher cover of avoiders than both extremes (Fig. 3a). Avoider cover increased with time since disturbance ($p = 0.0333$, Table 2a). Invader cover was higher in CC than in CL sites ($p = 0.0148$), which was in line with our hypothesis, but we found no impact of either season or TSD (Table 2b; Fig. 3b). There was a significant interaction between harvest type and season on endurer cover ($p = 0.0251$; Table 2c). Endurers were the most abundant in summer CL and the least abundant in summer CC, with intermediate values of cover for winter harvests, although a Tukey HSD test failed to find a significant treatment level difference ($p > 0.07$; Fig. 3c). While no statistical test was

Table 1
Permanova results for the plant community of the harvested sites.

	Df	Sum of squares	Mean squares	F value	R ²	p value
(a) Vascular plants only						
Harvest type	1	0.5153	0.5153	3.9175	0.126	0.0046
Season	1	0.0634	0.0634	0.4823	0.016	0.8202
Harvest type × Season	1	0.3377	0.3377	2.5673	0.083	0.0326
Time since disturbance	1	0.275	0.2750	2.0904	0.067	0.0646
Residuals	22	2.8937	0.1315		0.708	
Total	26	4.0850			1	
(b) Bryophytes and lichens only						
Harvest type	1	0.1986	0.1986	1.1828	0.043	0.2995
Season	1	0.1386	0.1386	0.8256	0.030	0.5703
Harvest type × Season	1	0.3685	0.3685	2.1954	0.080	0.0296
Time since disturbance	1	0.1948	0.1948	1.1606	0.042	0.3213
Residuals	22	3.693	0.1679		0.804	
Total	26	4.5935			1	

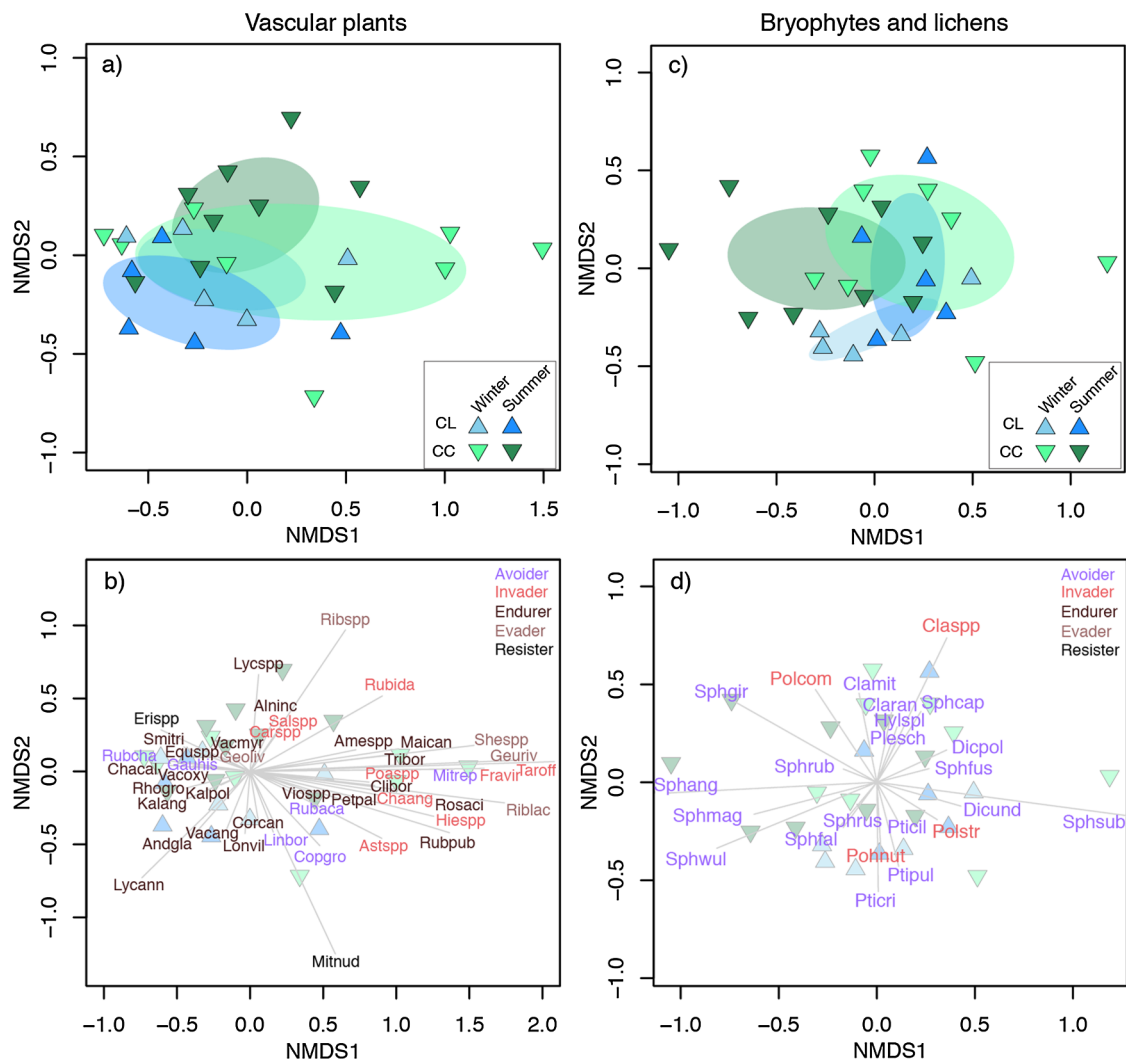


Fig. 2. Distribution of the harvested sites ($n = 27$) in two-dimensional NMDS ordinations (100 iterations, Bray–Curtis distance) based on vascular plant community composition (a–b, stress of 0.16, 45 taxa) and bryophytes and lichens (c–d, stress of 0.21, 23 taxa) community composition. Individual points represent sample units (sites) grouped by harvest type and season: careful logging (CL, blue), clear cuts (CC, green), winter (light colours), summer (dark colors). Ellipses represent the 95% confidence interval of each harvest type and season combinations (a, c). Species codes are provided in [Appendix B](#) and font colours indicate their functional type (b, e). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

conducted on evaders, they seemed to have a slightly higher cover in the CC treatments than in the CL treatments (Fig. 3d). Resisters, which only comprised *Eriophorum* spp., were uncommon in all the treatments types (Fig. 3e). Response of species by functional type to disturbance created by harvest operations was very variable (Appendix D). Individual avoider, endurers and invader species responded either to harvest type, or season, or both and were more abundant at intermediate or extreme levels of disturbance.

3.3. Comparison of community responses to harvest operations and fire severity

Disturbance severity, including fire types, had a significant impact on all the subsets of the understory communities investigated, explaining 36% of the variance for the entire vegetation communities, 31% for vascular plants only and 37% for the bryophytes and lichens only (Table 3). Since similar patterns were found for the three subsets of data, we focus our interpretation of results on the total vegetation community. Overall, community composition was similar among all harvest treatments, with only summer CL and summer CC being significantly different (Fig. 4a), but very distinct from that of both types of fires. Harvested sites tended to be more associated with species that were classified as invaders (e.g., *Taraxacum officinale*, *Chamerion*

angustifolium, and *Pohlia nutans*), while fire sites were dominated more heavily by avoiders (e.g., *Pleurozium schreberi*, *Hylocomium splendens*, *Cladonia stellaris*, and *Cladonia rangiferina*). Species from the endurer functional type were scattered among all disturbance types. No clear pattern was observed for evaders.

There was also an impact of TSD on all the subsets of the communities explored, explaining between 8 and 16% of the variance observed in community composition (Table 3). This impact of time since disturbance is apparent in the way that the sites separate in the multivariate space of the ordination, with fire sites being older than the harvested sites. Even if fire and harvest sites of all severities start with very different community compositions, our oldest harvested sites (~30 years old) seemed to converge towards a community composition similar to that of fire sites that were about 100 years old (Fig. 4c).

Each disturbance level was associated with at least one indicator species (Table 4). Overall, harvested sites were characterized by species from the invader functional group as well as from the pioneer evader group. One avoider, *Ptilidium pulcherrimum*, was an indicator species of winter CL sites. Resprouting endurers were identified as indicator species of summer CL and CC, as well as winter CC. Lastly, the two fire disturbances were characterized by late-successional avoider species of bryophytes (*Ptilium crista-castrensis* and *Pleurozium schreberi*) and

Table 2

Results from the linear models comparing functional type cover according to harvest type (CL vs. CC) and season (winter vs. summer), their interaction and time since disturbance (years). Non-significant interactions were removed from the models and type III sum of squares was used. The (+) indicates a significant positive effect of Time since disturbance.

	Sum of squares	Df	F value	p value
<i>(a) Avoiders</i>				
Harvest type	1600.0	1	6.1092	0.0213
Season	266.3	1	1.0169	0.3238
Time since disturbance (+)	1342.4	1	5.1295	0.0333
Residuals	5294.7	22		
<i>(b) Invaders</i>				
Harvest type	2177.0	1	6.9396	0.0148
Season	6.4	1	0.0204	0.8877
Time since disturbance	1269.8	1	3.5589	0.0561
Residuals	7215.3	23		
<i>(c) Endurers</i>				
Harvest type	890.9	1	6.6098	0.0174
Season	295.2	1	2.1906	0.1530
Harvest type × Season	778.8	1	5.7785	0.0251
Time since disturbance	4.2	1	0.0309	0.8622
Residuals	2965.2	22		

lichens (*Cladonia stellaris*; Table 4).

Contrary to our hypothesis, the cover of avoiders increased with increasing disturbance severity, possibly because of the age difference between harvested and fire origin sites (Table 5a; Fig. 5a and b). Indeed, we

found a marginally significant relationship between TSD and avoider cover. Even within each disturbance type, no clear pattern of disturbance severity could be detected (Table 5a; Fig. 5a). Invader cover was highest in winter CC sites, and lowest in the fire sites (Table 5b; Fig. 5c). All other harvest treatments showed invader covers that were not different from either winter CC or fire. There was no significant linear relationship between TSD and invader cover, although their cover was high in the harvested sites and decreased sharply after about 30 years post disturbance (Table 5; Fig. 5d). There was a small trend for endurer cover to be higher at the lower end of the soil disturbance severity gradient, but there was a significant interaction between disturbance severity and TSD (Table 5c; Fig. 5e). The highest covers were found in the CL and winter CC sites, intermediate covers were found in the summer CC and low severity fire sites, and the lowest cover was found in the high severity fire sites. When looking at individual disturbances' response to TSD, no clear pattern of change in terms of endurer cover emerged for any of the harvest types and seasons nor for low severity fires, with all having a relatively high and stable cover. However, in high severity fire sites, this functional group showed a clear increase in cover with TSD, with a low cover about 50 years post-disturbance that keeps increasing in the following years. Evaders seemed more common in fire than harvest sites (no statistical tests) and resisters had a low cover (< 3%) for all disturbance types and throughout TSD (Fig. 5).

3.4. Potential for paludification in harvest and fire origin sites

Sphagnum spp. recovery was faster in low than in high severity fires (Fig. 6a), but we found no impact of harvest type or season on *Sphagnum* cover (harvest type $F_{1,23} = 0.05$, $p = 0.8270$; season $F_{1,23} =$

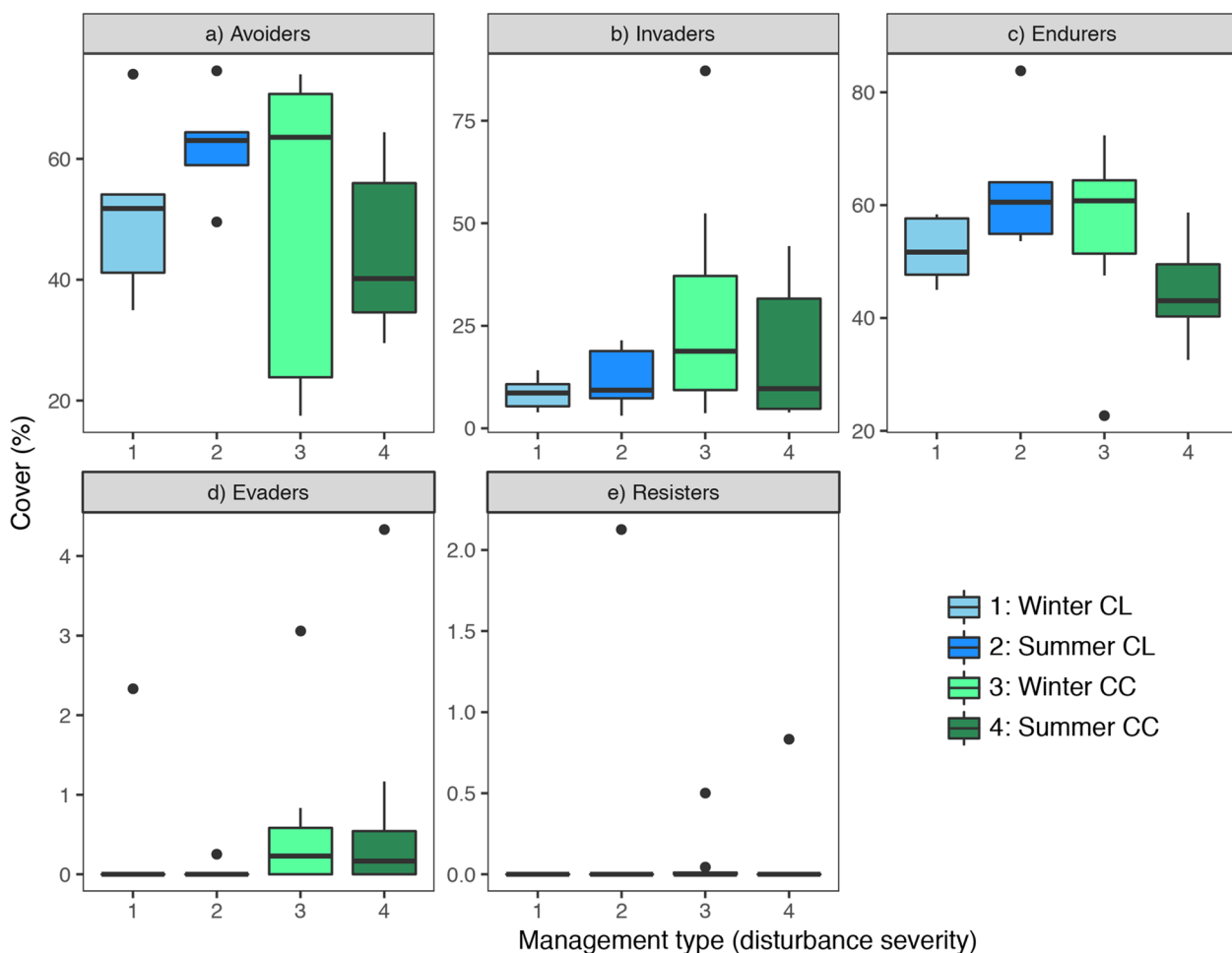


Fig. 3. Box and whiskers plots showing functional type cover according to harvest type (CL vs. CC) and season (winter vs. summer) and their interaction. The upper and lower hinges correspond to the first and third quartiles (the 25th and 75th percentiles). Outliers are shown as dots. Raw data is presented.

Table 3
Results of the perManova on Bray-Curtis transformed community data.

	DF	Sum of squares	Mean squares	F value	R ²	p value
<i>(a) All vegetation</i>						
Disturbance severity ¹	5	3.1263	0.62526	6.8061	0.35827	0.0004
Time since disturbance ²	1	1.1901	1.19009	12.9545	0.13638	0.0002
Residuals	48	4.4096	0.09187	0.50534		
Total	54	8.726	1			
<i>(b) Vascular plants only</i>						
Disturbance severity	5	3.7959	0.75917	4.9806	0.31307	< 0.0001
Time since disturbance	1	1.0123	1.01231	6.6413	0.08349	< 0.0001
Residuals	48	7.3164	0.15243	0.60344		
Total	54	12.1246	1			
<i>(c) Bryophytes and lichens only</i>						
Disturbance severity	5	2.4363	0.48726	7.4359	0.36729	0.0078
Time since disturbance	1	1.0515	1.0515	16.0467	0.15852	0.0002
Residuals	48	3.1453	0.06553	0.47419		
Total	54	6.6331	1			

¹ Disturbance severity was ordered from 1 to 6 in this order: winter CL, summer CL, winter CC, summer CC, low severity fire and high severity fire.

² Measured in years.

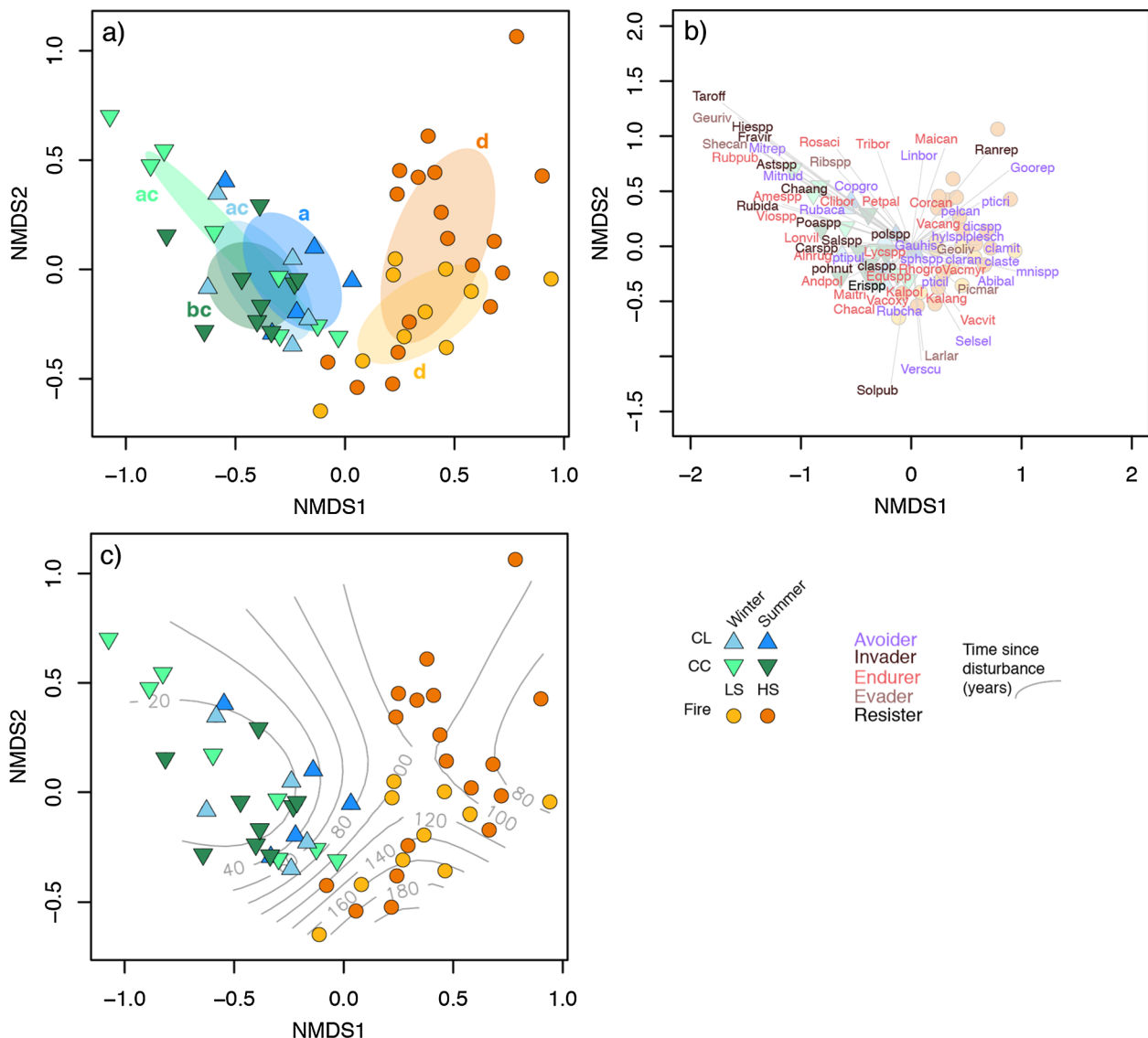


Fig. 4. Two-dimensional NMDS for fire and harvest data ($n = 55$ sites, stress 0.14). (a) Sites, disturbance type and 95% confidence interval ellipses. Different letters indicate that the groups were significantly different following a multivariate pairwise posthoc test. (b) Overlay of all species vectors. Text colors indicate functional groups. The list of species codes and attribution of a functional type is presented in [Appendix B](#). (c) Isolines representing time since disturbance (years). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4
Significant indicator species associated with each disturbance type.

Ordered factor	Disturbance type	Species	Functional type	Indicator value	p value
1	Winter CL	<i>Ptilidium pulcherrimum</i>	Avoider	0.4963	0.0054
		<i>Pohlia nutans</i>	Invader	0.4462	0.0086
2	Summer CL	<i>Vaccinium oxycoccos</i>	Endurer	0.3675	0.0086
		<i>Polytrichum</i> spp.	Invader	0.3257	0.0304
		<i>Rhododendron groenlandicum</i>	Endurer	0.2971	0.0054
3	Winter CC	<i>Clintonia borealis</i>	Endurer	0.4987	0.0086
		<i>Salix</i> spp.	Invader	0.4312	0.017
		<i>Rubus pubescens</i>	Endurer	0.4287	0.0104
		<i>Geum rivale</i>	Evader	0.375	0.0108
		<i>Shepherdia canadensis</i>	Evader	0.3396	0.0100
		<i>Fragaria virginiana</i>	Invader	0.3332	0.0336
		<i>Ribes</i> spp.	Evader	0.3173	0.0374
		<i>Taraxacum officinale</i>	Invader	0.25	0.031
		<i>Mitchella repens</i>	Avoider	0.2445	0.049
4	Summer CC	<i>Alnus incana</i> ssp. <i>rugosa</i>	Endurer	0.408	0.0090
5	Low severity fire	<i>Cladonia stellaris</i>	Avoider	0.57	0.0010
		<i>Picea mariana</i>	Evader	0.4982	0.0026
		<i>Kalmia angustifolia</i>	Endurer	0.4897	0.0068
6	High severity fire	<i>Ptilium crista-castrensis</i>	Avoider	0.5977	0.0170
		<i>Pleurozium schreberi</i>	Avoider	0.3139	0.0034

Table 5
Results from the models comparing functional type cover according to disturbance severity (1–6), time since disturbance (log transformed), and their interaction. The cover of invaders was square root transformed. Non-significant interactions were removed from the models and type III sum of squares was used.

	Sum of squares	Df	F value	p value
(a) Avoiders				
Disturbance severity	4556.4	5	4.9831	0.0009
Time since disturbance	647.9	1	3.5427	0.0659
Residuals	8777.9	48		
(b) Invaders				
Disturbance severity	33.619	5	3.4750	0.0092
Time since disturbance	0.865	1	0.4511	0.5050
Residuals	92.051	48		
(c) Endurers				
Disturbance severity	3091.2	5	5.3717	0.0006
Time since disturbance	224.6	1	1.9515	0.1696
Disturbance severity × Time since disturbance	1969.5	5	3.4224	0.0109
Residuals	4949	43		

0.69, $p = 0.4139$; harvest type × season $F_{1,23} = 0.61$, $p = 0.4410$). Harvested sites (TSD = 11–32 years) presented a cover of *Sphagnum* spp. that was more similar to much older post-fire sites, about 75 years old in low severity fires and about 150–200 years old in high severity fires (Fig. 6a). Harvest operations, regardless of the type or season, led to a community at a later stage of succession at an earlier age, at least for *Sphagnum* spp., which may increase the paludification rate. Additionally, we found that, regardless of the disturbance type or severity, an increase in ericaceous shrub cover was associated to an increase in *Sphagnum* spp. cover (Fig. 6b).

4. Discussion

In boreal ecosystems, understory communities are critical both in terms of biodiversity and of ecosystems processes, such as nutrient cycling, productivity, and soil microclimate regulation (Hart and Chen, 2006, 2008; Paquette and Messier, 2011; Turetsky et al., 2012). While the impacts of different harvest types and fires on tree regeneration, tree growth, and soil characteristics have been assessed (Lecomte et al., 2005, 2006a, Lafleur et al., 2010, 2016a), our study is the first one to comprehensively investigate these impacts on understory communities. We used data collected along a soil disturbance severity gradient by harvesting (CL and CC in winter and summer; Lafleur et al. (2010)) and high and low severity fires (Lecomte et al., 2005, 2006a) to assess how

these disturbances altered boreal plant communities in the Clay Belt area of eastern Canada. All subsets of understory communities investigated (vascular plants, bryophytes, and lichens) showed a strong response to disturbance type (CL vs. CC vs. fire), but little impact of harvest season or fire severity. Harvested understory communities reached a more advanced stage of succession faster, either through an accelerated rate or a more advanced start point. Indeed, these sites were all under 40 years post-harvest, but were more similar to 75–100 years fire-origin communities in composition and indeed if just *Sphagnum* spp. cover is examined mean post-harvest covers were similar to post-fire covers of 60 (low severity) or 150–200 (high severity) years post fire. Convergence of understory communities in sites originating from fire and harvest disturbances has also been reported in other parts of the boreal forest (Reich et al., 2001; Rees and Juday, 2002; Rydgren et al., 2004). Differences in community composition were reflected in the abundance of the various plant functional types, although individual species responses were idiosyncratic and likely depended on species' ecological tolerances. Species from the endurer functional type were the most common type in our plant communities, and were common across all disturbance types, severities and ages. This strategy seems to be successful in boreal ecosystems that are regularly affected by both large (fire) and small (windthrow or insect epidemics) disturbances (Payette, 1992; De Grandpré et al., 2003; Fenton et al., 2009). Our results support the strong resilience of coniferous boreal

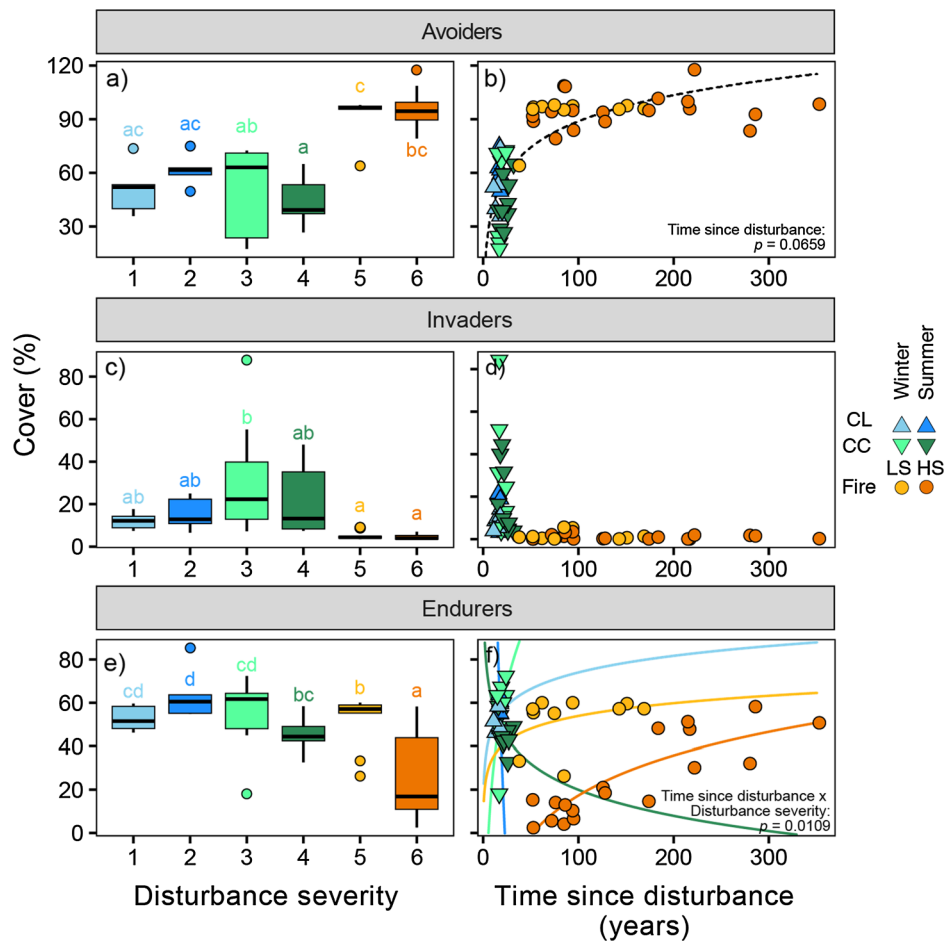


Fig. 5. Boxplot and scatter plots showing the relationship between the cover of the five plant functional types and disturbance severity (1: winter CL, 2: summer CL, 3: winter CC, 4: summer CC, 5: low severity fire, and 6 high severity fire) and time since disturbance (years). For invaders, the raw data is presented here, but analyses were run on square root transformed data. Different letters indicate that cover differed significantly between disturbances (Tukey HSD, $p < 0.05$). Lines indicate significant or near-significant relationships between time since fire and cover.

understory species, many of which possess traits and reproductive strategies that favour a fast reestablishment following disturbances (Rowe, 1983a,b; Reich et al., 2012; Hollingsworth et al., 2013).

4.1. Community responses to disturbance types and severities

In harvested sites, soil disturbances associated with the two types of logging considered affected understory vegetation communities (De

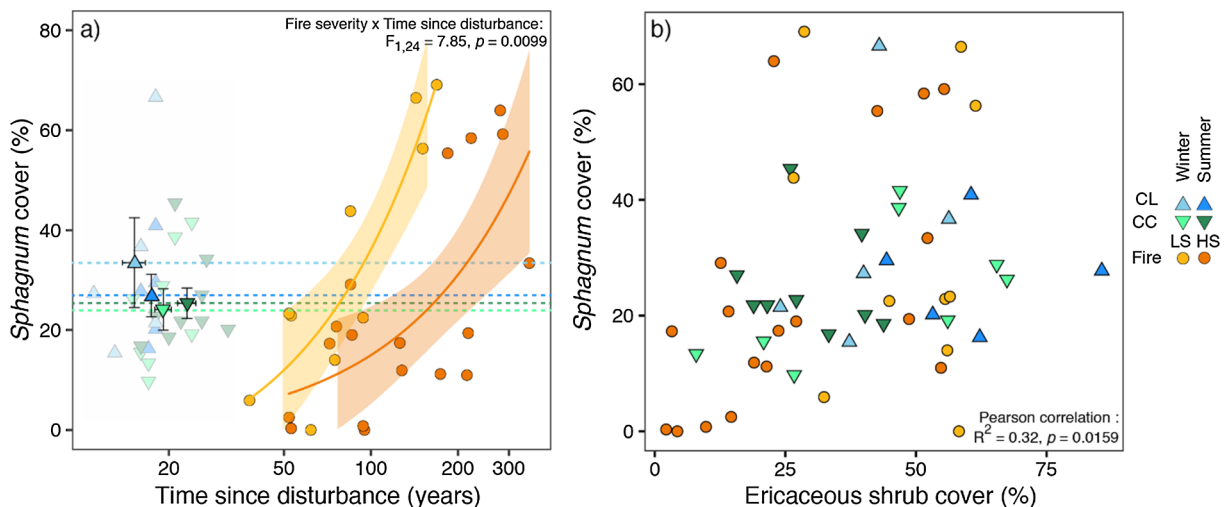


Fig. 6. (a) *Sphagnum* spp. cover (%) according to time since disturbance (years). Darker symbols with standard error bars and horizontal dashed lines represent the averages for each harvest type and season. Linear were model run on fire data only and linear relationships were plotted with the 95% confidence intervals (shaded areas). While time since disturbance was not transformed the analysis, we present a log scale in the plot to facilitate visualisation of the harvested sites. (b) *Sphagnum* spp. cover (%) according to ericaceous shrub cover (%).

Grandpré et al., 2003). Differences in vascular plant community composition were found between summer CL and summer CC, while the winter harvests produced communities that were intermediate. Because the frozen ground is less affected by machinery and plant communities are protected by the snow cover, it is possible that winter CC are more similar to CL harvests (Renard et al., 2016). For bryophytes and lichens, there was a trend for different communities in winter CL than in the other three more severe disturbances. Indeed, we found a higher cover of endurers, especially ericaceous shrubs, and avoiders in CL than CC harvests, as expected from lower levels of soil disturbances in the former. The hummock-forming *Sphagnum fuscum*, an avoider often associated with open peatlands, was more abundant in the open stands created by CL harvesting. Surprisingly, neither endurer species as a whole nor individual species showed a positive association with organic layer depth. Possibly, even the highest degree of mechanical soil disturbance associated with harvest was not sufficient to impede their resprouting capacity (De Grandpré et al., 2003). This was exemplified by *Alnus incana* subsp. *rugosa*, which was mostly found in summer CC sites. This species uses vegetative reproduction by stump sprouts; therefore, mechanical soil disturbances could stimulate its reproduction and growth. Avoider and endurer species not removed by harvest operations may have been able to withstand the increased light and altered soil moisture associated with post-harvest conditions or remained protected in areas where tree regeneration was left in place (Nguyen-Xuan et al., 2000; Paquette et al., 2016).

Soil disturbances associated with CC and summer harvests increase nutrient availability and lead to increased black spruce height and productivity (Lafleur et al., 2010, 2016a). We hypothesized that invader cover would be higher at sites affected by more severe soil disturbances, which was supported by our results with a higher cover of invaders in CC than CL sites. However, individual species responses did not follow our expectations, with *Cladonia* spp. being more common at intermediate disturbance severities and *Polytrichum strictum* being more abundant in CL sites, as this species tends to be found on organic soils and *Sphagnum* hummocks. Contrary to our expectations, we found a high cover of *Sphagnum girgensohnii* and *Sphagnum wulfianum*, avoider species associated with forested habitats, in summer CC sites, which we had posited as the most severe harvest disturbance. It is possible that the fast recovery of a tall black spruce canopy under this harvest method (Lafleur et al., 2010, 2016a) allowed for a fast recolonization or for the survival of remnant patches of these forest-associated species.

Large differences in community composition were found between harvest and fire sites, which point to different TSD between the two disturbance types, along with differences in soil disturbance severity. This pattern is usually attributed to the greater impact of fire on the forest floor compared to harvest operations (Lafleur et al., 2018). Harvest operations like CC or CL are mechanical disturbances that mostly affect soils by compaction, while fire is a chemical disturbance that consumes organic matter by oxidation, leading to a net loss in organic layer depth, and generates a large pulse of nutrients (Lafleur et al., 2016b, 2018). TSD is a critical driver of plant community composition, as species are associated to specific stand structures, light availabilities, and dispersal rates (Harper et al., 2005; Bergeron and Fenton, 2012; Fenton and Bergeron, 2013; Zouaoui et al., 2014; Lafleur et al., 2018). While our dataset did not include young fire-origin sites, with all of our fire sites being older than 38 years since fire, this gap in successional processes can be completed using the large body of work that has been completed comparing early post-fire and post-harvest plant communities in the boreal forest. Differences in community composition are usually found between harvest- and fire-origin sites in the first few years following disturbance (< 30 years old), with a higher abundance of pioneer species (invaders), such as *Marchantia polymorpha* L., *Polytrichum juniperinum* Hewd., *Polytrichum piliferum* Hewd., *Chamerion angustifolium*, and some graminoids, following fire (Noble et al., 1977; Nguyen-Xuan et al., 2000; Jean et al., 2017) and residual species from mature forests (avoiders) being more common following harvest (Brumelis and Carleton, 1989; Nguyen-Xuan et al., 2000; Rees and Juday, 2002; Paquette et al., 2016). Additionally, ecosystems are more resistant to

invading species when deeper residual organic layers and charred duff are found, given the poor quality of this type of substrate for generally small-seeded invaders (Kershaw, 1977; Walker et al., 2017). Since our sites are all liable to paludification, it is likely that even high severity fires expose little mineral soil, so initial post-disturbance succession and residual organic layer depths may differ from sites on drier substrates.

However, while we classified both high and low severity fires as the upper end of our disturbance severity gradient, they presented a lower cover of invaders than the harvested CC and CL sites where invader species such as *Taraxacum officinale*, *Chamerion angustifolium* and *Pohlia nutans* were more common. Indeed, pioneer species tend to be associated to high degrees of soil disturbance and exposed mineral soils and are therefore also commonly found in harvested stands that have experienced a high severity soil disturbance, such as CC (Noble et al., 1977; Outcalt and White, 1981; Carleton and MacLellan, 1994; Nguyen-Xuan et al., 2000). Harvest, compared to fire, would be expected to have a higher susceptibility of invasion as many of these species are exotics that are dispersed by human activities. Additionally, given that all of our fire sites were > 38 years old, it is likely that invaders would have been replaced by mid- to late-successional species after about 30 years post-disturbance (Jean et al., 2017). Even in our harvested sites, the cover of invaders was relatively low compared to avoiders or endurers, since this functional type tends to disappear with canopy closure as they are mostly shade-intolerant (De Grandpré et al., 2003). As expected, avoider cover increased with time since disturbance and was highest in older low and high severity fires, which had a high cover of bryophytes and lichens, such as *Pleurozium schreberi*, *Hylocomium splendens*, *Cladonia stellaris* and *Cladonia rangiferina*. Avoiders species, such as *Ptilidium pulcherrimum* or *Mitchella repens*, were also found in harvested stands (Brumelis and Carleton, 1989; Nguyen-Xuan et al., 2000; Rees and Juday, 2002; Paquette et al., 2016). All harvest treatments and low severity fires presented a high cover of endurer species with little variability over time, indicating that these species were either not fully removed by the disturbance, or that they had the capacity to resprout rapidly from their underground roots and rhizomes. In high severity fires, endurer abundance increased with time, which indicated that their underground roots and rhizomes were destroyed by combustion of the organic layer (Mallik, 2003; Fenton et al., 2005; Lafleur et al., 2016a). While CC reduced organic layer depth compared to CL, these differences were not significant, likely because most of the soil disturbances are limited to the skidding trails (Lafleur et al., 2016a). Aside from differences in development of the *Sphagnum* layer, plant communities of low and high severity fires were similar, since initial differences in species composition and organic layer depths would have had time to dissipate by the time the stands reached 50 years old (Lecomte et al., 2005; Renard et al., 2016).

4.2. Similarity between harvest and fire post-disturbance communities

We had hypothesized that differences in understory plant community among disturbance types and severities would become less pronounced over time, given the strong persistence and resilience of boreal species to disturbances (Rowe, 1983a,b; Reich et al., 2012; Hollingsworth et al., 2013). Nevertheless, in boreal forests of western Quebec, Nguyen-Xuan et al. (2000) and in eastern Québec Paquette et al. (2016) found no evidence of convergence in understory communities at sites originating from CC and fire after up to 23 years and 90 years post-disturbance, respectively. Conversely, we found that communities of about 30 years old harvested sites seem to converge towards fire-origin communities that were about 75–100 years old, which may indicate a different time-scale necessary to detect convergence. Additionally, we found that communities originating from CL disturbances seemed more similar to communities from fire-origin than communities from CC sites at a similar age. This indicates that succession of understory species may indeed reach a more advanced state at an early time since disturbance under lower soil disturbances with retaining of some of the tree cover and most of the soil and understory. Large-scale harvest operations only started in Abitibi around 1940 and the typical harvest rotation time is about

100 years (Cyr et al., 2009; Kuuluvainen and Gauthier, 2018). Future investigations will be required to truly assess the successional trajectories of stands recovering following harvest operations at a time scale that is comparable to the fire return interval in the study area (92–360 years; Bergeron et al. (2004)).

The Clay Belt of eastern Ontario and western Quebec is an area that is prone to paludification, given the poor drainage promoted by clay deposits and cold climate. Paludification leads to lower soil temperatures and nutrient availability (Heinselman, 1963; Van Cleve and Viereck, 1981); thus, lower stand productivity (Fenton et al., 2005; Lecomte et al., 2006a; Simard et al., 2007). In North American black spruce forests, establishment of *Sphagnum* carpets is common in 100 years old stands (Fenton and Bergeron, 2006; Turetsky et al., 2010). Our harvested sites (all < 40 years old) showed patterns of succession and *Sphagnum* covers that were similar to sites that were about 75–100 years in low severity fires or about 200 years in high severity fires. Unburned *Sphagnum* patches can serve as a refugia for recolonization following fire (Shetler et al., 2008), as evidenced from an apparent faster recovery of *Sphagnum* spp. cover in low than high severity fires (Lafleur et al., 2016a). In addition, residual *Sphagnum* patches may be common at harvested sites and *Sphagnum* can regenerate from vegetative fragments spread by mechanical disturbances (Rocheffort and Lode, 2006; Fenton and Bergeron, 2007). Ericaceous shrubs have roots and rhizomes buried deeply in the ground, which allows them to resprout quickly following disturbances that do not fully remove the organic layer (Mallik, 2003; Lafleur et al., 2016a; Renard et al., 2016). *Sphagnum* have been known to take advantage of the structure created by roots and stems of those shrubs to increase their vertical growth, which is known as the ladder effect (Malmer et al., 2003; Fenton and Bergeron, 2006). Indeed, *Sphagnum* cover was positively associated with ericaceous shrub cover. Therefore, soil disturbances associated with current logging practices may not be sufficient to prevent or slow paludification (Nguyen-Xuan et al., 2000; Lecomte et al., 2006a; Lachance et al., 2013). Increased soil disturbances associated with logging operations, such as the use of soil preparation techniques or prescribed burns, may be necessary in the Clay Belt area in order to maintain and restore forest productivity in the long-term (Lafleur et al., 2010, 2018; Faivre et al., 2016; Renard et al., 2016;

Morimoto and Juday, 2018).

5. Conclusion

In the Clay Belt area of eastern Canada, paludification is a common phenomenon in black spruce stands where commercial timber is harvested. In this study, we showed that differences in understory plant communities were found among harvest types (CL vs. CC), but that little change was observed according to harvest season. We had expected a strong response of understory species to the degree of soil disturbance associated with harvesting methods according to their functional types, with a higher cover of avoiders and endurers at the lower end of the soil disturbance gradient, and more invaders at sites where soils were more disturbed. This pattern was partially supported by our results, with variable species-specific responses. Lastly, we showed that high and low severity harvest methods did not result in post-disturbance understory plant communities that were similar to high and low severity fires. Indeed, young harvested stands showed community compositions and *Sphagnum* covers that were typical of much older fire-origin stands. Thus, it appears that the current methods employed with harvesting operations in the Clay Belt do not lead to a degree of soil disturbance that is sufficient to slow-down paludification, which may have implications for long-term forest productivity in the region (Lafleur et al., 2010; Faivre et al., 2016; Renard et al., 2016).

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Appendix A.: Average characteristics of the 27 harvested and 28 burned sites used in this study

	Season/ Severity	Severity index	Site code	Disturbance year ¹	Sampling year	Time since distur- bance ² (yr)	Organic layer depth (cm)	ROL depth ³ (cm)	Spruce density (stems ha ⁻¹)	Spruce height ⁴ (cm)
Careful log- ging (CL)	Winter	1	CN-24	1994	2007	13	46.1	–	2000	2.34
			WA-42	1989	2007	18	32.7	–	1000	2.4
			MI-56	1989	2007	18	121.2	–	1533	1.84
			MI-58	1991	2007	16	105.3	–	2267	1.8
			CN-70	1996	2007	11	77.0	–	1767	2.01
			CS-11	1990	2006	16	43.9	–	2667	2.19
	Summer	2	VP-35	1989	2007	18	121.1	–	2383	2.37
			VP-36	1989	2007	18	41.3	–	2317	2.26
			VP-37	1989	2007	18	58.2	–	1992	2.16
			WA-44	1994	2006	17	57.8	–	3592	2.08
			VP-33	1993	2006	13	40.6	–	1633	2.03
			VP-39	1989	2006	17	50.8	–	1650	2.03
			VP-40	1989	2006	17	51.7	–	1217	1.93
Clear cut (CC)	Winter	3	WA-43	1982	2006	24	76.5	–	4492	2.26
			MI-52	1987	2006	19	46.6	–	1792	2.20
			MI-53	1986	2007	21	36.9	–	2808	2.58
			MI-62	1991	2006	15	77.2	–	1775	1.73
			MI-64	1982	2006	24	39.3	–	2042	2.5
			CN-23	1975	2007	32	74.1	–	2700	3.38
			VP-32	1990	2006	16	60.2	–	1800	2.84
			WA-45	1993	2007	14	25.3	–	1908	2.65
			MI-54	1982	2008	26	62.4	–	2225	3.23
	Summer	4	MI-55	1982	2008	26	57.8	–	2967	3.05
			MS-63	1986	2007	21	78.8	–	2650	2.6
			MI-66	1986	2006	20	88.6	–	1683	2.56
			CN-68	1984	2006	22	51.1	–	1383	2.44
			CN-69	1980	2007	27	106.3	–	2683	2.71

Fire	Low	5	F3	1907	2000	94	50.8	16.2	–	–
			F5	1831	2000	169	44.6	8.4	–	–
			F11	1962	2000	38	27.2	12	–	–
			F17	1849	2000	151	52.7	17.3	–	–
			F53	1948	2002	53	21.1	7.3	–	–
			F58	1940	2002	62	20.9	5.8	–	–
			F60	1916	2002	85	35.1	6.7	–	–
			F63	1858	2001	143	40.7	7.9	–	–
		High	F74	1949	2001	52	35.8	13	–	–
			F78	1926	2001	75	45.4	17.9	–	–
			F2	1916	2000	85	19.1	1.9	–	–
			F4	1948	2000	53	11.8	2.5	–	–
			F6	1290	2000	280	47.7	3.2	–	–
			F7	1210	2000	217	37	1.7	–	–
			F8	1825	2000	174	29.7	1.4	–	–
			F9	730	2000	215	39	1.5	–	–
			F12	1905	2000	95	18.2	1.1	–	–
			F15	1924	2000	76	14.3	1.5	–	–
			F16	415	2000	286	58.9	2.7	–	–
			F18	1874	2000	126	23	2	–	–
			F20	– 355	2000	353	61	3.2	–	–
			F50	1635	2002	222	34.9	2	–	–
			F55	1907	2002	94	18.9	2.5	–	–
			F56	1916	2002	86	16.5	1.3	–	–
			F59	1930	2002	72	16.4	2.3	–	–
			F62	1949	2001	52	16.4	1.4	–	–
			F75	1873	2001	128	22.4	2	–	–
			F95	1817	2001	184	22.5	1.3	–	–

¹For fire sites, disturbance year was measured using ¹⁴C dating of charcoals in soils, but time since disturbance for these sites (Lecomte et al., 2006b).

²For fire sites, it was based on the age of the oldest tree found within the sites and sometimes differs from the ¹⁴C date (Lecomte et al., 2006b).

³ROL depth = residuals organic layer depth post disturbance. Only available for fire sites.

⁴Average black spruce height within the sites.

Appendix B.: Species list table

	Growth form	Family	Species name	Six-letter code	Reference
Avoiders					
Vascular plants	Tree	Pinaceae	<i>Abies balsamea</i> (L.) Mill.	Abibal	Rowe (1983a,b)
	Shrub (dwarf)	Ericaceae	<i>Gaultheria hispida</i> (L.) Muhl. ex Bigelow	Gauhis	Gilliam (2014)
		Caprifoliaceae	<i>Linnaea borealis</i> L.	Linbor	Rowe (1983a,b), Gilliam (2014)
		Rubiaceae	<i>Mitchella repens</i> L.	Mitrep	Coladonato (1993)
		Rosaceae	<i>Rubus arcticus</i> L. ssp. <i>acaulis</i> (Michx.) Focke	Rubaca	Nowak et al. (2002), Biswas and Mallik (2010)
		Rosaceae	<i>Rubus chamaemorus</i> L.	Rubcha	Gilliam (2014)
	Forb	Ranunculaceae	<i>Coptis groenlandica</i> (Oeder) Fernald	Copgro	Gilliam (2014)
	Forb	Orchidaceae	<i>Goodyera repens</i> (L.) R.Br.	Goorep	Rowe (1983a,b), Gilliam (2014)
	Forb	Saxifragaceae	<i>Mitella nuda</i> L.	Mitnud	Rowe (1983a,b), Gilliam (2014)
	Forb	Selaginellaceae	<i>Selaginella selaginoides</i> (L.) P. Beauv. ex Mart. & Schrank	Selsel	.
	Forb	Plantaginaceae	<i>Veronica scutellata</i> L.	Verscu	.
Lichen	Fruticose	Cladoniaceae	<i>Cladonia mitis</i> Sandst.	clamit	Nowak et al. (2002)
	Fruticose	Cladoniaceae	<i>Cladonia rangiferina</i> (L.) Weber ex F.H. Wigg.	claran	Nowak et al. (2002)
	Lichen	Cladoniaceae	<i>Cladonia stellaris</i> (Opiz) Pouzar & Vězda	claste	Gilliam (2014)
	Foliose	Peltigeraceae	<i>Peltigera canina</i> (L.) Willd.	pelcan	Rowe (1983a,b)
Bryophyte	Moss (acrocarpous)	Dicranaceae	<i>Dicranum</i> spp.	dicspp	.
	Moss (acrocarpous)	Dicranaceae	<i>Dicranum polysetum</i> Swartz	dicpol	.
	Moss (acrocarpous)	Dicranaceae	<i>Dicranum undulatum</i> Bridel	dicund	.
	Moss (acrocarpous)	Mniaceae	Mniaceae spp.	mnispp	Gilliam (2014)
	Moss (pleurocarpous)	Hylacomiaceae	<i>Hylocomium splendens</i> (Hedw.) Schimp.	hylspl	Rowe (1983a,b), Gilliam (2014)
	Moss (pleurocarpous)	Hylacomiaceae	<i>Pleurozium schreberi</i> (Brid.) Mitt.	plesch	Gilliam (2014)
	Moss (pleurocarpous)	Ptilidiaceae	<i>Ptilidium ciliare</i> (L.) Hampe	ptcil	Gilliam (2014)
	Moss (pleurocarpous)	Hypnaceae	<i>Ptilium crista-castrensis</i> (Hedw.) De Not.	pticri	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum</i> spp.	sphspp	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum angustifolium</i> (C.E.O. Jensen ex Russow) C.E.O. Jensen	sphang	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum capillifolium</i> (Ehrh.) Hedw.	sphcap	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum fallax</i> (H. Klinggr.) H. Klinggr.	sphfal	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum fuscum</i> (Schimp.) H. Klinggr.	sphfus	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum girgensohnii</i> Russow	sphgir	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum magellanicum</i> Brid.	sphmag	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum rubellum</i> Wilson	sphrub	Gilliam (2014)

	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum russowii</i> Warnst.	sphrus	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum subfulvum</i> Sjörs	sphsubful	Gilliam (2014)
	Moss (Sphagnum)	Sphagnaceae	<i>Sphagnum wulfianum</i> Girgensohn	sphwul	Gilliam (2014)
	Liverwort (leafy)	Ptilidiaceae	<i>Ptilidium pulcherrimum</i> (Weber) Vain.	ptipul	.
Endurers					
Vascular plants	Shrub	Betulaceae	<i>Alnus incana</i> subsp. <i>rugosa</i> (Du Roi) R. T. Clausen	Alnrug	Rowe (1983a,b), Gilliam (2014)
	Shrub	Rosaceae	<i>Amelanchier</i> spp.	Amespp	Gilliam (2014)
	Shrub	Ericaceae	<i>Andromeda polifolia</i> L.	Andpol	.
	Shrub	Ericaceae	<i>Chamaedaphne calyculata</i> (L.) Moench	Chacal	Gilliam (2014)
	Shrub	Ericaceae	<i>Kalmia angustifolia</i> L.	Kalang	Gilliam (2014)
	Shrub	Ericaceae	<i>Kalmia polifolia</i> Wangenh.	Kalpol	Gilliam (2014)
	Shrub	Caprifoliaceae	<i>Lonicera villosa</i> (Michx.) Schult.	Lonvil	Gilliam (2014)
	Shrub	Ericaceae	<i>Rhododendron groenlandicum</i> (Oeder) Kron & Judd	Rhogro	Rowe (1983a,b), Gilliam (2014)
	Shrub	Rosaceae	<i>Rosa acicularis</i> Lindl.	Rosaci	Nowak et al. (2002)
	Shrub	Ericaceae	<i>Vaccinium angustifolium</i> Aiton	Vacang	Gilliam (2014)
	Shrub	Ericaceae	<i>Vaccinium myrtilloides</i> Michx.	Vacmyr	Rowe (1983a,b), Gilliam (2014)
	Shrub (dwarf)	Rosaceae	<i>Rubus pubescens</i> Raf.	Rubpub	Gilliam (2014)
	Shrub (dwarf)	Ericaceae	<i>Vaccinium oxycoccos</i> L.	Vacox	Gilliam (2014)
	Shrub (dwarf)	Ericaceae	<i>Vaccinium vitis-idaea</i> L.	Vacvit	Rowe (1983a,b), Gilliam (2014)
	Forb	Liliaceae	<i>Clintonia borealis</i> (Aiton) Raf.	Clibor	Gilliam (2014)
	Forb	Cornaceae	<i>Cornus canadensis</i> L.	Corcan	Rowe (1983a,b), Gilliam (2014)
	Forb	Equisetaceae	<i>Equisetum</i> spp.	Equspp	Rowe (1983a,b), Gilliam (2014)
	Forb	Lycopodiaceae	<i>Lycopodium</i> spp.	Lycspp	Gilliam (2014)
	Forb	Lycopodiaceae	<i>Lycopodium annotinum</i> L.	Lycann	Gilliam (2014)
	Forb	Asparagaceae	<i>Maianthemum canadense</i> Desf.	Maican	Rowe (1983a,b), Gilliam (2014)
	Forb	Asteraceae	<i>Petasites frigidus</i> var. <i>palmatus</i> (L.) Fr. (Aiton) Cronquist	Petpal	Gilliam (2014)
	Forb	Asparagaceae	<i>Maianthemum trifolium</i> (L.) Sloboda	Maitri	Gilliam (2014)
	Forb	Primulaceae	<i>Trientalis borealis</i> Raf.	Tribor	Gilliam (2014)
	Forb	Violaceae	<i>Viola</i> spp.	Viospp	Gilliam (2014)
Invaders					
Vascular plants	Shrub	Rosaceae	<i>Rubus idaeus</i> L.	Rubida	Gilliam (2014)
	Shrub	Salicaceae	<i>Salix</i> spp.	Salssp	Gilliam (2014)
	Forb	Asteraceae	Asteraceae spp.	Astssp	Rowe (1983a,b), Gilliam (2014)
	Forb	Onagraceae	<i>Chamaerion angustifolium</i> (L.) Scop.	Chaang	Rowe (1983a,b), Gilliam (2014)
	Forb	Rosaceae	<i>Fragaria virginiana</i> Mill.	Fravir	Gilliam (2014)
	Forb	Asteraceae	<i>Hieracium</i> spp.	Hiespp	Gilliam (2014)
	Forb	Ranunculaceae	<i>Ranunculus repens</i> L.	Ranrep	USDA (2018)
	Graminoid	Cyperaceae	<i>Carex</i> spp.	Carspp	Rowe (1983a,b)
	Graminoid	Poaceae	Poaceae spp.	Poaspp	Rowe (1983a,b)
	Forb	Asteraceae	<i>Solidago puberula</i> Nutt.	Solpub	Gilliam (2014)
	Forb	Asteraceae	<i>Taraxacum officinale</i> (L.) Weber ex F.H. Wigg	Taroff	Royer and Dickinson (1999)
Lichen	Fruticose/ Squamulose	Cladoniaceae	<i>Cladonia</i> spp.	claspp	Gilliam (2014)
Bryophyte	Moss (acrocarpous)	Mniaceae	<i>Pohlia nutans</i> (Hedw.) Lindb.	pohnut	Barbé et al. (2017)
	Moss (acrocarpous)	Polytrichaceae	<i>Polytrichum</i> spp.	polssp	Rowe (1983a,b), Barbé et al. (2017)
	Moss (acrocarpous)	Polytrichaceae	<i>Polytrichum commune</i> Hedw.	polcom	Rowe (1983a,b), Barbé et al. (2017)
	Moss (acrocarpous)	Polytrichaceae	<i>Polytrichum strictum</i> Bridel, J. Bot. (Schrader)	polstr	Rowe (1983a,b), Barbé et al. (2017)
Evaders					
Vascular plants	Tree	Pinaceae	<i>Larix laricina</i> (Du Roi) K. Koch	Larlar	Uchytel (1991)
	Tree	Pinaceae	<i>Picea mariana</i> (Mill.) B.S.P.	Picmar	Rowe (1983a,b)
	Shrub	Grossulariaceae	<i>Ribes</i> spp.	Ribssp	Rowe (1983a,b), Gilliam (2014)
	Shrub	Grossulariaceae	<i>Ribes lacustre</i> (Pers.) Poir.	Riblac	Rowe (1983a,b), Gilliam (2014)
	Shrub	Elaeagnaceae	<i>Shepherdia canadensis</i> (L.) Nutt.	Shecan	Rowe (1983a,b)
	Forb	Santalaceae	<i>Geocaulon lividum</i> (Richardson) Fernald	Geoliv	Rowe (1983a,b)
	Forb	Rosaceae	<i>Geum rivale</i> L.	Geuriv	.
Resisters					
Vascular plant	Graminoid	Cyperaceae	<i>Eriophorum</i> spp.	Erispp	Rowe (1983a,b)

Appendix C.: Results of the perManova on Bray-Curtis transformed community data for the whole community ($n = 27$ sites, stress = 0.18)

	DF	Sum of squares	Mean squares	F value	R ²	p value
Harvest type	1	0.4008	0.40085	2.82177	0.096	0.0108
Season	1	0.08	0.08	0.56318	0.019	0.8138
Harvest type × Season	1	0.3351	0.33512	2.35908	0.080	0.0258
Time since disturbance	1	0.2467	0.24666	1.73638	0.059	0.0946
Residuals	22	3.1252	0.14206		0.746	
Total	26	4.1878			1	

Appendix D: Mean (S.E.) percent ground cover of understory species by harvest method and season, and plant response strategy to disturbance. Species needed to be present in at least 25% of the sites to be included, therefore no data is presented for evaders and resisters. *p = 0.1–0.051; **p = 0.05–0.001.; * p < 0.0001. Letters indicate that a Tukey HSD test was performed to assess treatment differences when a H(harvest) × S(season) interaction was found**

Species	CL ¹		CT ²		Harvest (H) ³	Season (S) ⁴	H × S ⁵	TSD ⁶
	Winter	Summer	Winter	Summer				
Avoider								
<i>Cladonia mitis</i>	0.05 (0.05)	0.31 (0.24)	0.24 (0.17)	0.27 (0.11)	−0.3	−0.92	0.72	0.67
<i>Cladonia rangiferina</i>	0.1 (0.06) a	1.32 (0.75) a	2.76 (1.07) a	1.2 (0.48) a	−0.83	−1.04	2.33*	1.77
<i>Coptis groenlandica</i>	0.22 (0.15)	0.35 (0.35)	0.03 (0.02)	0.1 (0.08)	−0.21	0.15	−0.45	−0.5
<i>Dicranum polysetum</i>	0.48 (0.2)	0.69 (0.24)	0.74 (0.33)	0.57 (0.25)	−0.91	−0.65	1.1	0.48
<i>Gaultheria hispidula</i>	2.79 (0.23)	6.49 (2.01)	3.24 (1.19)	2.93 (0.52)	−1.48	−1.76	1.12	−0.41
<i>Hylocomium splendens</i>	0.05 (0.04)	0.21 (0.16)	0.02 (0.01)	0.09 (0.03)	−0.12	−0.98	−0.19	−0.33
<i>Pleurozium schreberi</i>	9.8 (2.88) ab	22.55 (4.51) a	17.33 (4.9) ab	10.46 (2.81) b	−2.84*	−1.68	2.66*	1.71
<i>Ptilidium ciliare</i>	1.03 (0.38)	0.57 (0.14)	0.31 (0.21)	0.48 (0.15)	−0.35	0.57	−1.11	−0.12
<i>Ptilium crista-castrensis</i>	0.43 (0.11)	0.19 (0.13)	0.18 (0.08) ab	0.15 (0.09)	−1.36	2.24*	−0.99	2.37*
<i>Ptilidium pulcherrimum</i>	0.9 (0.58)	0.49 (0.23)	0.08 (0.05)	0.34 (0.2)	−1.61	1.32	−1.22	2.16*
<i>Sphagnum angustifolium</i>	0.14 (0.1)	0.09 (0.08)	0.05 (0.03)	1.03 (0.46)	1.77	0.18	−1.68	−0.07
<i>Sphagnum capillifolium</i>	5.1 (3.67)	4.83 (1.72)	6.84 (2.01)	2.21 (0.71)	−0.9	−0.34	1.25	0.03
<i>Sphagnum fallax</i>	12.38 (5.88)	11.09 (4.58)	4.92 (2.18)	7.28 (2.11)	−1.4	0.33	−0.58	1.45
<i>Sphagnum fuscum</i>	3.25 (2.41)	3.02 (1.69)	2.19 (1.1)	1.81 (0.74)	−0.45	−0.07	0.06	0.39
<i>Sphagnum girgensohnii</i>	0.17 (0.12) a	0.05 (0.04) a	0 (0) a	2.2 (0.93) b	3.42*	0	−2.81*	−1.62
<i>Sphagnum magellanicum</i>	5.65 (2.85)	1.64 (1.14)	4.55 (2.35)	6.65 (2.27)	1.03	1.32	−1.48	1.02
<i>Sphagnum rubellum</i>	2.14 (0.91)	1.02 (0.46)	1.53 (0.47)	2.1 (0.62)	0.43	0.84	−0.85	0.7
<i>Sphagnum russowii</i>	3.53 (1.08)	3.68 (2.24)	1.74 (1.29)	1.52 (0.49)	−1.14	0.52	−0.56	0.65
<i>Sphagnum subfulvum</i>	0.95 (0.92)	1.5 (1.5)	2.32 (1.67)	0.32 (0.19)	−0.33	−0.08	0.66	−0.05
<i>Sphagnum wulfianum</i>	0.19 (0.12) a	0.01 (0.01) a	0.01 (0.01) a	0.26 (0.18) a	0.77	1.61	−2.1*	1.07
Endurer								
<i>Alnus incana</i> subsp. <i>rugosa</i>	8.23 (4.53)	1.28 (0.7)	7.77 (2.8)	12.58 (2.97)	2.26*	1.39	−1.87	0.01
<i>Andromeda polifolia</i>	0.38 (0.31)	0.63 (0.39)	0.56 (0.31)	0.19 (0.14)	−2.02	0.01	1.3	2.34*
<i>Chamaedaphne calyculata</i>	8.72 (4.78)	4.67 (4.67)	3.91 (1.97)	4.82 (1.76)	−0.26	1.51	−0.97	2.09*
<i>Clintonia borealis</i>	0 (0)	0.05 (0.05)	0.71 (0.41)	0.2 (0.14)	1.02	−0.54	1.06	−1.04
<i>Cornus canadensis</i>	1.03 (0.51)	1.54 (0.66)	0.59 (0.24)	0.74 (0.39)	−0.62	−0.54	0.23	−0.54
<i>Equisetum</i> spp.	2.46 (0.45)	2.02 (0.65)	1.74 (0.36)	2.47 (0.72)	0.79	0.38	−0.98	−1.07
<i>Kalmia angustifolia</i>	0.58 (0.41)	1.23 (0.64)	2.27 (1.15)	0.74 (0.36)	−1.59	−0.42	1.67	1.87
<i>Kalmia polifolia</i>	0.17 (0.17)	0.12 (0.1)	0.05 (0.05)	0.08 (0.04)	−0.59	0.12	−0.29	1.15
<i>Lycopodium</i> spp.	0.41 (0.19) a	0 (0) a	0.16 (0.16) a	0.68 (0.39) a	1.82	1.63	−2.41*	0.2
<i>Maianthemum canadense</i>	0.11 (0.1)	0.1 (0.1)	0.18 (0.17)	0.06 (0.06)	0.15	0.07	0.03	−0.7
<i>Petasites frigidus</i> var. <i>palmatus</i>	1.24 (1.18)	0.33 (0.26)	0.59 (0.23)	0.09 (0.06)	−0.72	0.74	0.41	0.53
<i>Rhododendron groenlandicum</i>	20 (2.97)	33.67 (4.75)	18.9 (5.6)	9.78 (1.21)	−3.19*	−1.5	1.98	0.18
<i>Rubus pubescens</i>	0 (0)	0.25 (0.25)	1.59 (0.8)	0.01 (0.01)	−0.31	−0.62	2.06	−0.26
<i>Maianthemum trifolium</i>	1.3 (0.27)	1.29 (0.51)	1.34 (0.46)	1.83 (0.66)	0.25	0.31	−0.6	0.37
<i>Trientalis borealis</i>	0 (0)	0.06 (0.05)	0.15 (0.12)	0.03 (0.02)	0.02	−1.06	1.25	−0.68
<i>Vaccinium angustifolium</i>	0.92 (0.47)	5.43 (1.75)	2.19 (1.47)	3.01 (1.66)	−1.47	−1.86	1.21	0.33
<i>Vaccinium myrtilloides</i>	5.08 (1.2)	6.77 (1.4)	10.52 (2.1)	6.87 (1.53)	1.15	−1.22	1.3	−2.54*
<i>Vaccinium oxycoccos</i>	1.48 (0.43)	2.2 (0.49)	0.63 (0.19)	1.16 (0.24)	−1.49	−1.05	−0.13	0.2
<i>Viola</i> spp.	0.41 (0.26)	0.06 (0.06)	0.34 (0.15)	0.22 (0.1)	0.65	1.14	−0.47	0.22
Invader								
<i>Asteraceae</i> spp.	0.17 (0.11)	0.36 (0.36)	0.39 (0.26)	0.03 (0.03)	−0.66	−0.08	0.95	−0.23
<i>Carex</i> spp.	2.5 (1.14)	2.53 (1.16)	4.94 (1.65)	5.05 (2.05)	1.18	−0.09	−0.13	−0.93
<i>Chamaerion angustifolium</i>	0.58 (0.24)	0.47 (0.45)	1.26 (0.74)	0.26 (0.14)	0.44	0.47	−0.07	−0.99
<i>Cladonia</i> spp.	0 (0) a	0.28 (0.18) a	0.18 (0.09) a	0 (0) a	−1.2	−2.32*	2.95*	−1.45
<i>Poaceae</i> spp.	0.96 (0.4)	0.93 (0.71)	5.89 (2.66)	2.43 (1.32)	1.47	−0.11	0.29	−1.77
<i>Pohlia nutans</i>	0.92 (0.35)	0.63 (0.32)	0.12 (0.06)	0.39 (0.19)	−1.25	1.16	−1.35	1.04
<i>Polytrichum commune</i>	0.04 (0.04)	0.84 (0.73)	0.56 (0.32)	0.8 (0.44)	0.81	−1.45	0.4	−1.02
<i>Polytrichum strictum</i>	2.19 (0.54)	3.11 (0.65)	2.21 (0.6)	0.96 (0.26)	−2.35*	−0.9	1.71	−0.16
<i>Rubus idaeus</i>	0.11 (0.11)	0.1 (0.1)	1.21 (0.75)	1.37 (1.06)	1.94	−0.35	−0.28	−1.94
<i>Salix</i> spp.	0.93 (0.51)	2.61 (1.3)	8.3 (2.88)	6.7 (2.44)	1.8	−1.02	0.75	−1.13

¹ Careful logging.² Clearcutting.³ Harvest (CT).⁴ Season (winter).⁵ Harvest (CT) × Season (winter).⁶ Time since disturbance (years).

Appendix E.: Individual species responses to environmental variables

Species	Time since disturbance	Base saturation	CN ratio	Organic layer depth	P	pH	Black spruce density ¹	Black spruce height
(a) Avoider								
<i>Cladonia mitis</i>	0.5597	0.0681	0.1153	0.8285	0.9945	0.4441	0.5231	0.8718
<i>Cladonia rangiferina</i>	0.0268 (+)	0.0023 (+)	0.0396 (+)	0.3682	0.1232	0.9726	0.2312	0.2093
<i>Coptis groenlandica</i>	0.981	0.0307 (–)	0.2911	0.4197	0.8948	0.3269	0.6468	0.6651
<i>Dicranum polysetum</i>	0.8046	0.6545	0.0023 (+)	0.4964	0.4066	0.4646	0.2124	0.5416
<i>Gaultheria hispida</i>	0.1583	0.2773	0.6091	0.2843	0.2468	0.4589	0.0223 (+)	0.7018
<i>Hylocomium splendens</i>	0.1732	0.542	0.8446	0.5725	0.8997	0.9579	0.1611	0.1784
<i>Pleurozium schreberi</i>	0.4087	0.4972	0.0265 (+)	0.0945	0.4109	0.3458	0.0933	0.2822
<i>Ptilidium ciliare</i>	0.2957	0.0797	0.6116	0.178	0.9021	0.5494	0.5855	0.7429
<i>Ptilium crista-castrensis</i>	0.4861	0.0035 (–)	0.7892	0.3652	0.4112	0.3307	0.7794	0.4715
<i>Ptilidium pulcherrimum</i>	0.8081	0.0004 (–)	0.006 (+)	0.7716	0.642	0.2802	0.8715	0.5073
<i>Sphagnum angustifolium</i>	0.8331	0.8769	0.116	0.3555	0.4941	0.716	0.9352	0.4888
<i>Sphagnum capillifolium</i>	0.559	0.5123	0.314	0.2137	0.9968	0.9951	0.7728	0.2341
<i>Sphagnum fallax</i>	0.879	0.0174 (–)	0.0786	0.0103 (+)	0.5947	0.0849	0.4642	0.8998
<i>Sphagnum fuscum</i>	0.411	0.3229	0.0839	0.1018	0.6404	0.351	0.385	0.4335
<i>Sphagnum girgensohnii</i>	0.0184 (–)	0.118	0.4771	0.9291	0.4606	0.5256	0.4485	0.0004 (+)
<i>Sphagnum magellanicum</i>	0.7718	0.6736	0.061	0.1681	0.7125	0.0965	0.915	0.695
<i>Sphagnum rubellum</i>	0.5642	0.9249	0.5452	0.9892	0.5542	0.4455	0.4009	0.9677
<i>Sphagnum russowii</i>	0.2483	0.3539	0.9553	0.7789	0.7415	0.8215	0.0462 (+)	0.6791
<i>Sphagnum subfulvum</i>	0.3016	0.4741	0.4616	0.431	0.42	0.2132	0.4995	0.211
<i>Sphagnum wulfianum</i>	0.4788	0.6409	0.8206	0.9969	0.7189	0.9767	0.4008	0.061
(b) Endurer								
<i>Alnus incana</i> subsp. <i>rugosa</i>	0.404	0.6343	0.0079 (–)	0.1089	0.2965	0.9165	0.6623	0.0393 (+)
<i>Andromeda polifolia</i>	0.008 (+)	0.1405	0.938	0.7808	0.6221	0.0843	0.3011	0.0108 (–)
<i>Chamaedaphne calyculata</i>	0.1101	0.2764	0.676	> 0.0001 (+)	0.4617	0.9558	0.613	0.3545
<i>Clintonia borealis</i>	0.6289	0.049 (+)	0.9395	0.1669	0.617	0.5757	0.2264	0.9038
<i>Cornus canadensis</i>	0.7439	0.0100 (–)	0.4712	0.0016 (–)	0.1545	0.4381	0.9443	0.7329
<i>Equisetum</i> spp.	0.3329	0.5855	0.6514	0.7565	0.3611	0.4909	0.3302	0.5752
<i>Kalmia angustifolia</i>	0.2873	0.3064	0.278	0.8368	0.2322	0.2601	0.4106	0.4822
<i>Kalmia polifolia</i>	0.5549	0.2916	0.1665	0.8175	0.3993	0.0678	0.0462 (+)	0.3875
<i>Lycopodium</i> spp.	0.0633	0.2203	0.1362	0.7136	0.6858	0.7726	0.7553	0.0048 (+)
<i>Maianthemum canadense</i>	0.7657	0.9741	0.5719	0.295	0.4166	0.2249	0.5604	0.922
<i>Petasites frigidus</i> var. <i>pal-matus</i>	0.0954	0.0214 (–)	0.7932	0.0006 (–)	0.663	0.0063 (+)	0.2425	0.1169
<i>Rhododendron groenlandicum</i>	0.7709	0.041 (–)	0.1387	0.4943	0.4641	0.0139 (–)	0.4856	0.1104
<i>Rubus pubescens</i>	0.1769	0.1653	0.7482	0.1071	0.7502	0.0198 (+)	0.5501	0.1839
<i>Maianthemum trifolium</i>	0.6248	0.4407	0.8694	0.0331 (+)	0.0282 (+)	0.491	0.9933	0.9462
<i>Trientalis borealis</i>	0.9021	0.285	0.9267	0.2014	0.1586	0.6655	0.5515	0.8951
<i>Vaccinium angustifolium</i>	0.9107	0.1707	0.9409	0.1553	0.3184	0.4845	0.519	0.7832
<i>Vaccinium myrtilloides</i>	0.1653	0.0008 (+)	0.1444	0.9005	0.9659	0.5985	0.9295	0.4401
<i>Vaccinium oxycoccus</i>	0.5704	0.8245	0.5038	0.0587	0.6353	0.2617	0.552	0.8976
<i>Viola</i> spp.	0.1863	0.2373	0.0238 (–)	0.0822	0.2274	0.6244	0.0545	0.4176
(c) Invader								
<i>Asteraceae</i> spp.	0.1767	0.2653	0.4071	0.0741	0.5069	0.4053	0.2595	0.0731
<i>Carex</i> spp.	0.6356	0.0406 (+)	0.3145	0.8038	0.7896	0.3641	0.1131	0.8531
<i>Chamaerion angustifolium</i>	0.7679	0.7772	0.3514	0.2907	0.587	0.0004 (+)	0.4699	0.8758
<i>Cladonia</i> spp.	0.9167	0.0045 (+)	0.1044	0.2572	0.7151	0.7816	0.1565	0.1256
<i>Poaceae</i> spp.	0.7166	0.2283	0.452	0.4872	0.9705	0.0372 (+)	0.2824	0.6969
<i>Pohlia nutans</i>	0.5958	0.0031 (–)	0.3054	0.172	0.5599	0.6801	0.8085	0.779
<i>Polytrichum commune</i>	0.3918	0.2076	0.5444	0.6513	0.1147	0.2906	0.8627	0.13
<i>Polytrichum strictum</i>	0.7336	0.0809	0.5006	0.7097	0.5183	0.3336	0.2331	0.0553
<i>Rubus idaeus</i>	0.2746	0.0017 (+)	0.7638	0.8091	0.2681	0.0682	0.1235	0.1411
<i>Salix</i> spp.	0.946	0.0059 (+)	0.6756	0.9047	0.3695	0.1305	0.8504	0.2896

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